1BP461 – Homework assignment 1 – 2026

Summary

The homework task is to:

1. Apply unconstrained and constrained mean-variance portfolio optimization to selected ETFs
2. Extend the portfolio optimization to a set of dual-momentum ETF allocation strategies

Requirements

The deadline for the delivery is **15.12.2026**. The solutions should be **uploaded** to the **InSIS repository** specially created for this purpose. The homework should be done in **R**.

IMPORTANT: Your uploaded solutions should contain the following parts:

A. **Short report in MS Word or PDF:** Describing the individual steps of the analysis and their main results with short discussion and interpretation of the results.

B. **Codes in R:** The R codes that you used to solve the homework.

Both parts **A.** and **B.** should be submitted to successfully complete the homework.

Part 1 – Mean-variance portfolio optimization on factor ETFs

Summary

The goal of Part 1 of the case study is to apply mean-variance portfolio optimization to a dataset of 6 factor ETFs, offering exposure to the Value and the Size factor. Four portfolios should constructed: (A) Equally-weighted, (B) Minimum-variance, (C) Tangency-portfolio, and a (D) Constrained-Tangency-portfolio (described later). The portfolio optimization should be run on an in-sample dataset (2007/05-2023/12) and its performance assessed on an out-sample dataset (2024/01-2026/10).

Dataset

The optimization should be run on the following 8 ETFs:

1. **SPY** = SPDR S&P 500 ETF Trust (US equities)
2. **EFA** = iShares MSCI EAFE ETF (International equities)
3. **LQD** = iShares Inv Grade Corporate Bond ETF (Investment grade bonds)
4. **HYG** = iShares High Yield Corporate Bond ETF (High yield bonds)
5. **VNQ** = Vanguard Real Estate Index Fund ETF (Real estate)
6. **REM** = iShares Mortgage Real Estate ETF (Mortgage real estate)
7. **GLD** = SPDR Gold Trust (Gold)
8. **TLT** = iShares 20+ Year Treasury Bond ETF (US Treasury bonds)

For the application of the active allocation strategies (in Part 2) you will need also:

i. **BIL** = SPDR Bloomberg 1-3 Month T-Bill ETF (US Treasury bills)

With the data downloaded for the period since the introduction of these ETFs up to 2024/10 from the Yahoo Finance website with the Tidyquant package function tq_get.

Portfolios

Using the in-sample part of the dataset (2007/05-2023/12), the following portfolios should be defined:

A. **Equally weighted portfolio:** Where all weights are set equal to $\boldsymbol{w} = \mathbf{1}/n_{Assets}$

B. **Minimum variance portfolio:** Find analytically so that $\boldsymbol{w} \propto \boldsymbol{\Sigma}^{-1}\mathbf{1}$
C. **Tangency portfolio:** Maximum Sharpe ratio portfolio, i.e. $\boldsymbol{w} \propto \boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu} - r_f\mathbf{1})$
D. **Constrained tangency portfolio:** Maximum Sharpe ratio portfolio found numerically with constrain that $0 < \boldsymbol{w} < 0.5$ (i.e. long-only portfolio with a maximum weight 50% per asset).

For simplicity, the risk-free rate should be set to $r_f = 3\%$ p.a. (divided by 12 to get the p.m. value).

Steps

The R code should contain the following steps:

1. Download OHLCV data for the 8 ETFs in the period from 2007/05-2026/10
2. Compute daily logarithmic returns of all ETFs
3. Compute the monthly simple returns of all ETFs
4. Convert the monthly returns in period 2007/05-2023/12 into a matrix
5. Compute the mean vector $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Sigma}$
6. Find the weights of portfolios A., B., C. and D.
7. Compute the realized out-sample returns (2024/01-2026/10)
8. Plot chart with the cumulative out-sample returns
9. Compute average out-sample returns and realized Sharpe ratios

Outputs

The following outputs should be included in the prepared document:

i. Weights of the optimized portfolios A., B., C., D. computed on 2007/05-2023/12
ii. Mean-returns, volatilities and Sharpe ratios used in the optimization on 2007/05-2023/12
iii. Chart with the in-sample cumulative returns of all portfolios on 2007/05-2023/12
iv. Chart with the out-sample cumulative returns of all portfolios on 2024/01-2026/10
v. Out-sample mean returns, volatilities and Sharpe ratios of all portfolios on 2024/01-2026/10
vi. Discussion and interpretation of the outputs

Part 2 – Portfolio optimization of dynamic ETF allocation strategies

Summary

The goal of Part 2 is to define a set of four dual-momentum dynamic ETF allocation strategies based on the 8 ETFs from Part 1. The strategies are based on the paper *"Risk Premia Harvesting Through Dual Momentum"* (Antonacci, 2017). Once the allocation strategies are defined, the portfolio optimization from Part 1 should be applied to the strategy returns to find their optimal weights.

Dual momentum strategies

The general framework of the dual momentum active allocation strategies is to compare at the end of each month, the 12-month momentum (i.e. the return over the previous 12 months) of (i) ETF X, (ii) ETF Y, and (iii) the BIL ETF, and then invest into either (i), (ii) or (iii), depending on which of the three ETFs (i.e. X, Y or BIL) has the highest return. The procedure is repeated at the end of each month.

Depending on the selected ETFs X and Y, four strategies are defined:

i. **Equity dual momentum:** Selects between US Equities (SPY) and International Equities (EFA)
ii. **Credit dual momentum:** Selects between investment grade bonds (LQD) and High-yield bonds (HYG).
iii. **Real estate dual momentum:** Selects between equity REITs (VNQ) and mortgage REITs (REM)
iv. **Economic stress dual momentum:** Selects between gold (GLD) and US Treasuries (TLT)

I.e. the “Equity dual momentum” selects at the end of each month between an investment into SPY, EFA and BIL, based on their 12-month momentum. The “Credit dual momentum” selects similarly between LQD, HYG and BIL, etc.

Steps

The R code should contain the following steps:

1. Download OHLCV data for the 9 ETFs in the period from 2007/05-2026/10
2. Compute daily logarithmic returns of all ETFs
3. Compute the monthly simple returns of all ETFs
4. Compute returns of the four dual momentum strategies (Equity, Credit, Real estate and Stress)
5. Convert the monthly strategy returns in period 2007/05-2023/12 into a matrix
6. Compute the mean vector $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Sigma}$ (of the strategy returns)
7. Find the weights of portfolios A., B., C. and D.
8. Compute the realized out-sample returns (2024/01-2026/10)
9. Plot chart with the cumulative out-sample returns
10. Compute average out-sample returns and realized Sharpe ratios